Akshay Kumar

Kanpur, Uttar Pradesh, India | +91 8958222898 | akshaykr7775@gmail.com | Google Scholar | LinkedIn | Website

SUMMARY

Ph.D. in Mechanical Engineering from IIT Kanpur with expertise in Finite Element Analysis (FEA), vibration analysis, and experimental validation. Experienced in computational modelling of complex material behaviour, including hyperelasticity, viscoelasticity, and geometric nonlinearity. Strong research background in solid mechanics, structural dynamics, machine learning, and computer vision, with publications in peer-reviewed international journals and conference proceedings.

RESEARCH INTERESTS

Finite Element Analysis; Vibration Analysis; Solid Mechanics; Machine Learning

CORE SKILLS

FEA & Simulation: ABAQUS, HyperMesh, NASTRAN, PATRAN, COMSOL, AutoCAD, SolidWorks

Programming: Python, MATLAB

Experimental Characterization: Experimental Modal Analysis (EMA) using accelerometers, impact hammer, Digital Image Correlation (DIC) using high-speed cameras.

EDUCATION

M.Tech-Ph.D., Mechanical Engineering - Indian Institute of Technology Kanpur Aug 2018 - Dec 2026*

Specialization: Solid Mechanics & Design · CPI: 8.0/10

Ph.D. Thesis: Experimental and Waveguide-Based Finite Element Investigations for Analyzing Vibrations in Tyres with Anisotropy, Hyperelasticity, Viscoelasticity, and Geometric Nonlinearity.

M.Tech Thesis: Classification of Tabla Bols in Indian Music using Deep Learning Models.

Advisor: Prof. Nachiketa Tiwari

*Expected PhD defence; thesis submitted in July 2026

B.Tech, Mechanical Engineering - Meerut Institute of Engineering and Technology 2012 - 2016
71.4%

EXPERIENCE

FARE Fellow (Postdoctoral Research) - Indian Institute of Technology Kanpur Sep 2026 - Present

- Currently extending the WFEM framework to incorporate air-cavity and tyre-rotation effects. Also working on a Physics-Informed Neural Network (PINN) for a vibration problem.

FEA Project - Submarine Project (sponsored by Mazagon Dock Shipbuilders Ltd.) March 2025 - May 2025
Indian Institute of Technology Kanpur · PI: Prof. Nachiketa Tiwari

- Delivered structural and vibration FEA modeling and simulation as part of a funded, timeline-bound R&D project.

Machine Learning Intern - TSTS Pvt. Ltd., Kanpur March 2025 - Aug 2025

- Built computer vision and anomaly-detection models for railway applications.

JOURNAL PUBLICATIONS

- **Akshay Kumar**, Madhavrao Londhe, Chirag Patel, Nachiketa Tiwari *"An Experimentally Validated Waveguide-Based Finite Element Model for a Tyre with Hyperelastic, Viscoelastic and Anisotropic Effects."* Journal of Vibration Engineering & Technologies (June 2026). DOI: 10.1007/s42417-026-02628-z

CONFERENCE PROCEEDINGS

- **Akshay Kumar**, Sreerag Ashok, Nachiketa Tiwari. *"Indian music tabla bols classification using deep learning."* 188th Meeting of the Acoustical Society of America, Louisiana, USA (May 2025). DOI: 10.1121/2.0002191
- Anubhav Mishra, **Akshay Kumar,** Nachiketa Tiwari, *"Neural Network and Kinematic Control of a Stewart Platform for Animatronic Application,"* 6th International Conference on Recent Advances in Information Technology (RAIT), Dhanbad, India, 2025, DOI: 10.1109/RAIT65068.2025.11089023.

TEACHING EXPERIENCE

Tutor at IIT Kanpur

- Engineering Graphics (TA111, UG, 3 times) 2023–2026

Teaching Assistant at NPTEL Swayam Courses 2022–2026

- Basics of Finite Element Analysis-I & II (3 times each)
- Noise Management & Control
- Introduction to Composites

Teaching Assistant at IIT Kanpur 2019–2022

- Engineering Acoustics and Its Control (ME778a, PG)
- Vibration & Control (ME354a, UG)

PROFESSIONAL WORK

1. Reviewer for journals: Journal of Vibration Engineering & Technologies (Springer, 3 times) 2026
2. Completed the Finlatics Data Science with Python Program Dec 2024 - Feb 2025
3. Conducted lectures on Signal Processing for HAL Employees at IIT Kanpur. 2023-2024
4. Attended a four-week technical training at the Army Base workshop, Meerut, and NTPC, Delhi. 2015

POSITION OF RESPONSIBILITY

1. Senator, Student Gymkhana, IIT Kanpur 2019-2020
2. Sports Secretary, Hall 8, IIT Kanpur 2019-2020

REFERENCES

1. **Dr. Nachiketa Tiwari** (M.Tech - Ph.D. Thesis supervisor)
   Professor, Department of Mechanical Engineering,
   Indian Institute of Technology Kanpur, India
   Email: ntiwari@iitk.ac.in
2. **Dr. Akhilesh Mimani**
   Associate Professor, Department of Mechanical Engineering,
   Indian Institute of Technology Kanpur, India
   Email: amimani@iitk.ac.in
3. **Dr. Anubhav Mishra**
   Assistant Professor, Department of Design
   Indian Institute of Technology Guwahati, India
   Email: anubhv@iitg.ac.in